

Concept 2 | Trigonometric Identities and Expression Evaluation

English Student Edition • Focused formulas and guided practice

Formula opener

Identity

$$\sin^2 \theta + \cos^2 \theta = 1$$

Tangent

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Secant

$$1 + \tan^2 \theta = \sec^2 \theta$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q001 Written

Given (a) $\sin \theta = -\frac{1}{5}$ in QIII and (b) $\tan \theta = -\sqrt{6}$ with $\frac{3\pi}{2} < \theta < 2\pi$, find the other two trigonometric values in each case.

Solution

1. Use $\sin^2 \theta + \cos^2 \theta = 1$, then choose the sign from the quadrant.
2. Use $\tan \theta = \sin \theta / \cos \theta$ (or $1 + \tan^2 \theta = 1 / \cos^2 \theta$).

Final answer

$$(a) \cos \theta = -\frac{2\sqrt{6}}{5}, \tan \theta = \frac{\sqrt{6}}{12}; \quad (b) \cos \theta = \frac{\sqrt{7}}{7}, \sin \theta = -\frac{\sqrt{42}}{7}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q002 Written

Given one value and the quadrant, find the other two: (a) $\cos \theta = -1/\sqrt{3}$, QIII; (b) $\tan \theta = -2$, QIV; (c) $\sin \theta = 3/5$, QII; (d) $\tan \theta = \sqrt{3}/2$, $\pi < \theta < 3\pi/2$.

Solution

1. Find the missing squared value from a Pythagorean identity.
2. Select signs using the stated quadrant, then form the remaining ratio.

Final answer

$$(a) \sin \theta = -\frac{\sqrt{6}}{3}, \tan \theta = \sqrt{2}; (b) \cos \theta = \frac{\sqrt{5}}{5}, \sin \theta = -\frac{2\sqrt{5}}{5}; (c) \cos \theta = -\frac{4}{5}, \tan \theta = -\frac{3}{4}; (d) \sin \theta = -\frac{\sqrt{21}}{7}, \cos \theta = -\frac{2\sqrt{7}}{7}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q003 Written

For each of the twelve numbered cases in the given Exercise, one of $\sin \theta$, $\cos \theta$, $\tan \theta$ and a quadrant/interval is given. Find the other two values.

Solution

1. Square the known ratio and use $\sin^2 \theta + \cos^2 \theta = 1$ or $1 + \tan^2 \theta = 1/\cos^2 \theta$.
2. Apply the quadrant sign before simplifying the final ratio.

Final answer

$$(a) \left(-\frac{\sqrt{7}}{4}, \frac{3\sqrt{7}}{7}\right) (b) \left(-\frac{\sqrt{5}}{5}, -\frac{1}{2}\right) (c) \left(\frac{\sqrt{2}}{4}, -\frac{\sqrt{14}}{4}\right) (d) \left(-\frac{\sqrt{5}}{3}, -\frac{2\sqrt{5}}{5}\right) (e) \left(-\frac{\sqrt{15}}{4}, -\sqrt{15}\right) (f) \left(\frac{1}{3}, -\frac{2\sqrt{2}}{3}\right) (g) \left(-\frac{3\sqrt{10}}{10}, \frac{\sqrt{10}}{10}\right) (h) \left(-\frac{\sqrt{5}}{5}, -2\right) (i) \left(-\frac{12}{13}, -\frac{12}{5}\right) (j) \left(\frac{12}{13}, -\frac{5}{12}\right) (k) \left(-\frac{3}{5}, \frac{3}{4}\right) (l) \left(\frac{2\sqrt{5}}{5}, -\frac{\sqrt{5}}{5}\right)$$

Q004 Written

Using trigonometric identities, find the other two values given $\tan \theta + \cot \theta = 12$ and $\pi < \theta < 2\pi$.

Solution

1. Use $\tan \theta + \cot \theta = (\sin^2 \theta + \cos^2 \theta)/(\sin \theta \cos \theta)$ to obtain $\sin \theta \cos \theta = 1/12$.
2. The interval gives QIII or QIV; take the pair consistent with the stated signs.

Final answer

$$\sin \theta = -\frac{1}{2}, \quad \cos \theta = -\frac{\sqrt{3}}{2} \quad \text{or the swapped pair}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q005 Written

Given $\sin \theta = \frac{5}{13}$ **in QII, find** $\cos \theta$ **and** $\tan \theta$.

Solution

1. Use $\cos^2 \theta = 1 - 25/169$ and QII signs.

Final answer

$$-\frac{12}{13}, -\frac{5}{12}$$

Q006 Written

Given $\cos \theta = -\frac{3}{5}$ **in QIII, find** $\sin \theta$ **and** $\tan \theta$.

Solution

1. Use the Pythagorean identity and QIII signs.

Final answer

$$-\frac{4}{5}, \frac{4}{3}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q007 Written

Given $\tan \theta = \frac{3}{4}$ **in QI, find** $\sin \theta$ **and** $\cos \theta$.

Solution

1. Use a 3-4-5 triangle with positive signs.

Final answer

$$\frac{3}{5}, \frac{4}{5}$$

Q008 Written

Given $\tan \theta = -\frac{5}{12}$ **in QIV, find** $\sin \theta$ **and** $\cos \theta$.

Solution

1. Use a 5-12-13 triangle and QIV signs.

Final answer

$$-\frac{5}{13}, \frac{12}{13}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q009 Written

Given $\sin \theta = -\frac{\sqrt{2}}{2}$ **in QIII, find** $\cos \theta$ **and** $\tan \theta$.

Solution

1. The reference angle is $\pi/4$; both sine and cosine are negative in QIII.

Final answer

$$-\frac{\sqrt{2}}{2}, 1$$

Q010 MCQ

If $\sin \theta = -\frac{3}{5}$ **in QIII, $\cos \theta$ is:**

Solution

1. QIII makes cosine negative.

Correct option: B

Final answer

$$-\frac{4}{5}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q011 MCQ

If $\tan \theta = 2$ in QI, $\sin \theta$ is:**Solution**

1. Use a 1-2- $\sqrt{5}$ triangle.

Correct option: A

Final answer

$$\frac{2\sqrt{5}}{5}$$

Q012 MCQ

If $\cos \theta = \frac{1}{\sqrt{2}}$ in QIV, $\tan \theta$ is:**Solution**

1. The reference angle is $\pi/4$ and tangent is negative in QIV.

Correct option: B

Final answer

$$-1$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q013 MCQ

For $\tan \theta = -\sqrt{3}$ in QII, $\sin \theta$ is:

Solution

1. Reference angle $\pi/3$, sine positive in QII.

Correct option: C

Final answer

$$\frac{\sqrt{3}}{2}$$

Q014 MCQ

If $\sin \theta = \frac{7}{25}$ in QI, $\tan \theta$ is:

Solution

1. The triangle is 7-24-25.

Correct option: A

Final answer

$$\frac{7}{24}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q015 Written

Given $\sin \theta + \cos \theta = \frac{1}{2}$, **with** θ **in QII**, **evaluate** (a) $\sin \theta \cos \theta$, (b) $\sin^3 \theta + \cos^3 \theta$, (c) $\sin \theta - \cos \theta$.

Solution

1. Square the sum: $1 + 2 \sin \theta \cos \theta = 1/4$.
2. Use $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$, then choose the QII sign for (c).

Final answer

$$(a) -\frac{3}{8} \quad (b) \frac{11}{16} \quad (c) \frac{\sqrt{7}}{2}$$

Q016 Written

Given $\sin \theta + \cos \theta = \frac{1}{3}$, **with** θ **in QII**, **evaluate** (a) $\sin \theta \cos \theta$, (b) $\sin^3 \theta + \cos^3 \theta$, (c) $\sin \theta - \cos \theta$.

Solution

1. Square the given sum to find the product.
2. Use the sum-of-cubes identity and the quadrant sign.

Final answer

$$(a) -\frac{4}{9} \quad (b) \frac{37}{27} \quad (c) \frac{\sqrt{17}}{3}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q017 Written

Solve the three numbered Exercise problems: (1) given $\sin \theta - \cos \theta = \sqrt{3}/2$ in QI; (2) given $\sin \theta + \cos \theta = \sqrt{3}/2$ in QIV; (3) given $\sin \theta - \cos \theta = 1/2$ in QIII. Evaluate the three expressions requested in each case.

Solution

1. Square the given sum or difference to obtain the product.
2. Use the sum/difference of cubes and select the sign from the quadrant.

Final answer

$$(1) \sin \theta \cos \theta = \frac{1}{8}, \sin^3 \theta - \cos^3 \theta = \frac{7\sqrt{3}}{16}, \sin \theta + \cos \theta = \frac{\sqrt{5}}{2}; (2) -\frac{1}{8}, \frac{9\sqrt{3}}{16}, -\frac{\sqrt{5}}{2}; (3) \frac{3}{8}, \frac{11}{16}, -\frac{\sqrt{7}}{2}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q018 Written

Given $\sin \theta - \cos \theta = 0.5$ and θ in QIII, evaluate the exact value of $\sin^6 \theta - \cos^6 \theta$.

Solution

1. Find $\sin \theta \cos \theta = 3/8$, then $\sin^3 \theta - \cos^3 \theta = 11/16$ and $\sin^3 \theta + \cos^3 \theta = -5\sqrt{7}/16$.
2. Multiply the last two expressions.

Final answer

$$-\frac{55\sqrt{7}}{256}$$

Q019 Written

Given $\sin \theta + \cos \theta = \frac{2}{3}$ in QII, find $\sin \theta \cos \theta$ and $\sin \theta - \cos \theta$.

Solution

1. Square the sum and use QII signs.

Final answer

$$-\frac{5}{18}, \frac{\sqrt{14}}{3}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q020 Written

Given $\sin \theta - \cos \theta = \frac{\sqrt{5}}{2}$ **in QI, find** $\sin \theta \cos \theta$ **and** $\sin^3 \theta - \cos^3 \theta$.

Solution

1. Use $(s - c)^2 = 1 - 2sc$, then the difference-of-cubes identity.

Final answer

$$-\frac{1}{8}, \frac{7\sqrt{5}}{16}$$

Q021 Written

Given $\sin \theta + \cos \theta = -\frac{1}{2}$ **in QIII, find** $\sin^3 \theta + \cos^3 \theta$.

Solution

1. Find $sc = -3/8$, then apply the sum-of-cubes identity.

Final answer

$$-\frac{11}{16}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q022 Written

Given $\sin \theta - \cos \theta = \frac{1}{3}$ **in QIV, find** $\sin \theta + \cos \theta$.

Solution

1. Find $sc = 4/9$, so the square of the sum is $17/9$; use QIV sign.

Final answer

$$\frac{\sqrt{11}}{3}$$

Q023 Written

Given $\sin \theta - \cos \theta = 1$ **in QIII, find** $\sin^3 \theta - \cos^3 \theta$.

Solution

1. The product is zero and the difference-of-cubes factor is $1 + sc$.

Final answer

$$\frac{3}{2}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q024 MCQ

If $s + c = \frac{1}{2}$, then sc equals:

Solution

1. Square the sum and use $s^2 + c^2 = 1$.

Correct option: A

Final answer

$$-\frac{3}{8}$$

Q025 MCQ

If $s + c = \frac{1}{2}$, then $s^3 + c^3$ equals:

Solution

1. Use $(s + c)^3 - 3sc(s + c)$.

Correct option: B

Final answer

$$\frac{11}{16}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q026 MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s + c$ equals:

Solution

1. The sum is negative in QIII.

Correct option: B

Final answer

$$-\frac{\sqrt{7}}{2}$$

Q027 MCQ

If $s - c = \frac{\sqrt{3}}{2}$ in QI, then sc equals:

Solution

1. Use $(s - c)^2 = 1 - 2sc$.

Correct option: A

Final answer

$$\frac{1}{8}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q028 MCQ

If $s - c = 0.5$ in QIII, then $s^6 - c^6$ equals:

Solution

1. Use the two cubic factors and multiply.

Correct option: A

Final answer

$$-\frac{55\sqrt{7}}{256}$$

Q029 Written

Prove $\frac{\sin \theta}{1 + \cos \theta} + \frac{1}{\tan \theta} = \frac{1}{\sin \theta}$.

Solution

1. Rewrite $1/\tan \theta = \cos \theta/\sin \theta$.
2. Use the common denominator $(1 + \cos \theta)\sin \theta$, then $\sin^2 \theta + \cos^2 \theta = 1$.

Final answer

$$\text{LHS} = \text{RHS}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q030 Written

Prove (a) $\frac{1-\tan \theta}{1+\tan \theta} = \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta}$; **(b)** $\frac{\cos \theta}{1-\sin \theta} - \tan \theta = \frac{1}{\cos \theta}$.

Solution

1. For (a), multiply numerator and denominator by $\cos \theta$.
2. For (b), multiply the first fraction by $(1 + \sin \theta)/(1 + \sin \theta)$, then simplify.

Final answer

Both identities verified

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q031 Written

Prove the six identities in the given Exercise: (a) $\tan^2 \theta - \sin^2 \theta = \tan^2 \theta \sin^2 \theta$; (b) $\tan \theta + 1/\tan \theta = 1/(\sin \theta \cos \theta)$; (c) $\cos \theta/(1 - \sin \theta) - 1/\cos \theta = \tan \theta$; (d) $\tan \theta/\sin \theta - \sin \theta/\tan \theta = \sin \theta \tan \theta$; (e) $(1 + \sin \theta)/\cos \theta + \cos \theta/(1 + \sin \theta) = 2/\cos \theta$; (f) $(\tan^2 \theta - \sin^2 \theta)/\sin^2 \theta = \tan^2 \theta$.

Solution

1. Replace tangent by $\sin \theta / \cos \theta$ where useful.
2. Combine fractions and use $\sin^2 \theta + \cos^2 \theta = 1$.

Final answer

LHS = RHS for (a)+(f)

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q032 Written

Prove that, for all defined values of x , $\frac{1+\sin x-\cos x}{1+\sin x+\cos x} + \frac{1+\sin x+\cos x}{1+\sin x-\cos x} = 2 \csc x$.

Solution

1. Set $A = 1 + \sin(x)$ and $(B = \cos(x))$, combine the two fractions, and simplify the numerator to $(2(1 + \sin(x)))$.
2. Use $(1 - \sin^2 x = \cos^2 x)$ in the denominator to obtain $(2/\sin(x))$.

Final answer

$$\text{LHS} = 2 \csc x$$

Q033 Written

Prove $\frac{1+\sin \theta}{\cos \theta} = \frac{\cos \theta}{1-\sin \theta}$.

Solution

1. Cross-multiply and use $1 - \sin^2 \theta = \cos^2 \theta$.

Final answer

Identity proved

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q034 **Written**

Prove $\frac{1}{1+\cos \theta} + \frac{1}{1-\cos \theta} = 2 \sec^2 \theta.$

Solution

1. Combine over $1 - \cos^2 \theta = \sin^2 \theta$, then rewrite.

Final answer

Identity proved

Q035 **Written**

Prove $\frac{\tan \theta}{1+\tan \theta} = \frac{\sin \theta}{\sin \theta + \cos \theta}.$

Solution

1. Multiply numerator and denominator on the left by $\cos \theta$.

Final answer

Identity proved

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q036 Written

Prove $\frac{\sin \theta}{1 - \cos \theta} = \frac{1 + \cos \theta}{\sin \theta}.$

Solution

1. Multiply top and bottom by $1 + \cos \theta$, then use the Pythagorean identity.

Final answer

Identity proved

Q037 Written

Prove $\frac{\cos \theta}{1 + \sin \theta} = \frac{1 - \sin \theta}{\cos \theta}.$

Solution

1. Rationalise the left denominator and cancel $\cos^2 \theta$.

Final answer

Identity proved

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q038 MCQ

The first move in proving a trig identity is usually to:**Solution**

1. Keep one side fixed and transform the other.

Correct option: B**Final answer**

Simplify one side

Q039 MCQ

The identity used to replace $1 - \sin^2 \theta$ is:**Solution**

1. From $\sin^2 \theta + \cos^2 \theta = 1$.

Correct option: A**Final answer** $\cos^2 \theta$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q040 MCQ

$\tan \theta$ should be rewritten as:

Solution

1. Use the quotient identity.

Correct option: B

Final answer $\sin \theta / \cos \theta$

Q041 MCQ

To add two trig fractions, first find a:

Solution

1. Fraction addition requires a common denominator.

Correct option: B

Final answer

common denominator

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q042 MCQ

The result of a valid identity proof is:**Solution**

1. The two expressions are shown equal.

Correct option: B**Final answer**

LHS=RHS

Q043 MCQ

If $\cos \theta = -\frac{12}{13}$ in QII, $\sin \theta$ is:**Solution**

1. Use the 5-12-13 triangle and the QII sign.

Correct option: A**Final answer** $\frac{5}{13}$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q044 MCQ

If $\tan \theta = -\frac{1}{2}$ in QII, $\cos \theta$ is:

Solution

1. Cosine is negative in QII.

Correct option: A

Final answer

$$-\frac{2\sqrt{5}}{5}$$

Q045 MCQ

If $\sin \theta = \frac{8}{17}$ in QIV, $\tan \theta$ is:

Solution

1. The 8-15-17 triangle has negative sine in QIV.

Correct option: B

Final answer

$$-\frac{8}{15}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q046 MCQ

If $\tan \theta = \sqrt{3}$ and θ is in QIII, $\sin \theta$ is:

Solution

1. Reference angle $\pi/3$, with both ratios negative in QIII.

Correct option: D

Final answer

$$-\frac{\sqrt{3}}{2}$$

Q047 MCQ

If $s + c = 1$, then sc equals:

Solution

1. Square the sum and use $s^2 + c^2 = 1$.

Correct option: A

Final answer

$$0$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q048 MCQ

If $s - c = \sqrt{2}$, then sc equals:

Solution

1. Use $(s - c)^2 = 1 - 2sc$.

Correct option: A

Final answer

$$-\frac{1}{2}$$

Q049 MCQ

If $s + c = \frac{2}{3}$, then $s^3 + c^3$ equals:

Solution

1. Use $(s + c)^3 - 3sc(s + c)$.

Correct option: A

Final answer

$$\frac{19}{27}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q050 MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s^3 - c^3$ equals:

Solution

1. Use $(s - c)(1 + sc)$ with $sc = 3/8$.

Correct option: A

Final answer

$$\frac{11}{16}$$

Q051 MCQ

Which identity converts $\frac{1}{\tan \theta}$ into sine and cosine?

Solution

1. Take the reciprocal of tangent.

Correct option: B

Final answer

$$\frac{\cos \theta}{\sin \theta}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q052 MCQ

The identity $1 - \cos^2 \theta$ simplifies to:**Solution**

1. Rearrange the Pythagorean identity.

Correct option: A**Final answer**

$$\sin^2 \theta$$

Q053 MCQ

When proving an identity, which side should normally be transformed?**Solution**

1. Keep one side fixed and simplify the other to it.

Correct option: B**Final answer**

One side only

Worked Solutions

Trigonometric Identities and Expression Evaluation

Q054 MCQ

After combining trig fractions, the next step is usually to use:

Solution

1. Replace $1 - \sin^2\theta$ or $1 - \cos^2\theta$ using the identity.

Correct option: B

Final answer

The Pythagorean identity

Quiz MCQ 1 Concept Quiz · MCQ

If $\sin\theta + \cos\theta = \frac{1}{2}$, then $\sin\theta \cos\theta$ is:

Solution

1. Square the sum and use $s^2 + c^2 = 1$.

Correct option: A

Final answer

$-\frac{3}{8}$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Quiz MCQ 2 [Concept Quiz · MCQ](#)

If $\tan \theta = \frac{3}{4}$ in QI, then $\cos \theta$ is:

Solution

1. Use a 3-4-5 triangle.

Correct option: B

Final answer

$$\frac{4}{5}$$

Quiz MCQ 3 [Concept Quiz · MCQ](#)

Which identity is always valid where defined?

Solution

1. Use the fundamental quotient identity.

Correct option: B

Final answer

$$1 + \tan^2 \theta = \frac{1}{\cos^2 \theta}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Quiz MCQ 4 **Concept Quiz · MCQ**

If $s - c = \frac{1}{2}$ in QIII, then $s + c$ is:

Solution

1. Square the difference and use the QIII sign.

Correct option: B

Final answer

$$-\frac{\sqrt{7}}{2}$$

Quiz WRITTEN 5 **Concept Quiz · Written**

Given $\cos \theta = -\frac{5}{13}$ in QIII, find $\sin \theta$ and $\tan \theta$.

Solution

1. Use the 5-12-13 triangle and QIII signs.

Final answer

$$-\frac{12}{13}, \frac{12}{5}$$

Worked Solutions

Trigonometric Identities and Expression Evaluation

Quiz WRITTEN 6 Concept Quiz · Written

Given $s + c = \frac{2}{3}$, find sc and $s^3 + c^3$.

Solution

1. Square the sum to find sc , then apply the sum-of-cubes identity.

Final answer

$$-\frac{5}{18}, \frac{19}{27}$$

Quiz WRITTEN 7 Concept Quiz · Written

Prove $\frac{1-\sin \theta}{\cos \theta} = \frac{\cos \theta}{1+\sin \theta}$.

Solution

1. Cross-multiply and use $1 - \sin^2 \theta = \cos^2 \theta$.

Final answer

Identity proved

Worked Solutions

Trigonometric Identities and Expression Evaluation

Quiz WRITTEN 8 Concept Quiz · Written

Given $\tan \theta = -2$ **in QIV, find** $\sin \theta$ **and** $\cos \theta$, **then verify** $1 + \tan^2 \theta = 1/\cos^2 \theta$.

Solution

1. Use a 1-2- $\sqrt{5}$ triangle and QIV signs.
2. Substitute into both sides of the identity.

Final answer

$$\sin \theta = -\frac{2\sqrt{5}}{5}, \cos \theta = \frac{\sqrt{5}}{5}; 6 = 6$$

Use the question number shown on each page.